Chapter 2. Differentiation

The derivative, continuously differentiable functions, the chain rule, the inverse and implicit function theorems

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Overview

We define the derivative of a function $f : \mathbb{R}^m \to \mathbb{R}^n$ as a matrix that gives the best linear approximation to the increment $f(\mathbf{a}+\mathbf{h}) - f(\mathbf{a})$. The two major results are the *inverse function theorem*, which gives conditions under which a differentiable map $\mathbb{R}^n \to \mathbb{R}^n$ has a differentiable inverse, and the *implicit function theorem*, the theoretical basis of implicit differentiation. Elements of $\mathbb{R}^m$ and $\mathbb{R}^n$ are column matrices; $|\cdot|$ is the sup norm (Chapter 1).

§5. The Derivative

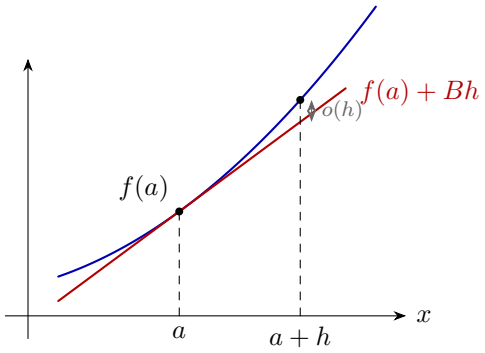


Figure. *The derivative $Df(\mathbf{a}) = B$ is the linear map for which $f(\mathbf{a}+\mathbf{h}) - f(\mathbf{a}) - B\mathbf{h}$ is small compared with $|\mathbf{h}|$: the graph of f is approximated near $\mathbf{a}$ by the affine map $\mathbf{h} \mapsto f(\mathbf{a}) + B\mathbf{h}$.*

For $\phi : A \to \mathbb{R}$ with $A \subset \mathbb{R}$ containing a neighborhood of a, the derivative is $\phi'(a) = \lim_{t\to 0}(\phi(a+t) - \phi(a))/t$. From it one gets (1) differentiable functions are continuous, (2) composites of differentiable functions are differentiable. For $f : \mathbb{R}^m \to \mathbb{R}^n$ we cannot divide by a vector, so we must reformulate.

Definition 5.1 (directional derivative)

Let $A \subset \mathbb{R}^m$, $f : A \to \mathbb{R}^n$, and suppose A contains a neighborhood of $\mathbf{a}$. Given $\mathbf{u} \in \mathbb{R}^m$, $\mathbf{u} \neq \mathbf{0}$, define

$$f'(\mathbf{a};\mathbf{u}) = \lim_{t\to 0} \frac{f(\mathbf{a}+t\mathbf{u}) - f(\mathbf{a})}{t},$$

provided the limit exists. It is the **directional derivative** of f at $\mathbf{a}$ with respect to $\mathbf{u}$ ($\mathbf{u}$ need not be a unit vector).

Example 5.1

Let $f : \mathbb{R}^2 \to \mathbb{R}$, $f(x_1, x_2) = x_1 x_2$, and $\mathbf{a} = (a_1, a_2)$. For $\mathbf{u} = (1, 0)$, $f'(\mathbf{a};\mathbf{u}) = \lim_{t\to 0} \frac{(a_1+t)a_2 - a_1 a_2}{t} = a_2$; for $\mathbf{v} = (1, 2)$, $f'(\mathbf{a};\mathbf{v}) = \lim_{t\to 0} \frac{(a_1+t)(a_2+2t) - a_1 a_2}{t} = a_2 + 2a_1$.

The existence of all directional derivatives is *not* a good definition of differentiability: it does not imply continuity (Example 5.3), nor that composites are differentiable.

Reformulate the one-variable case: ϕ is differentiable at a iff there is a number λ with

$$\frac{\phi(a+t) - \phi(a) - \lambda t}{t} \to 0 \quad \text{as } t \to 0;$$

then $\lambda = \phi'(a)$, and the linear function λt is the "first-order" (linear) approximation to the increment $\phi(a+t) - \phi(a)$. This generalizes.

Definition 5.2 (derivative)

Let $A \subset \mathbb{R}^m$, $f : A \to \mathbb{R}^n$, and suppose A contains a neighborhood of $\mathbf{a}$. f is **differentiable** at $\mathbf{a}$ if there is an n by m matrix B such that

$$\frac{f(\mathbf{a}+\mathbf{h}) - f(\mathbf{a}) - B \cdot \mathbf{h}}{|\mathbf{h}|} \to \mathbf{0} \quad \text{as } \mathbf{h} \to \mathbf{0}.$$

The matrix B is unique; it is the **derivative** of f at $\mathbf{a}$, denoted $Df(\mathbf{a})$.

The quotient is defined for $\mathbf{h}$ in a deleted neighborhood of $\mathbf{0}$. Using $\|\mathbf{h}\|$ instead of $|\mathbf{h}|$ gives an equivalent definition. *Uniqueness*: if C also satisfies the condition, subtracting gives $(C-B)\cdot\mathbf{h}/|\mathbf{h}| \to \mathbf{0}$. Fix $\mathbf{u}$, put $\mathbf{h} = t\mathbf{u}$ and let $t \to 0$: $(C-B)\cdot\mathbf{u}/|\mathbf{u}| = \pm(C-B)\cdot t\mathbf{u}/|t\mathbf{u}| \to \mathbf{0}$, so $(C-B)\cdot\mathbf{u} = \mathbf{0}$ for all $\mathbf{u}$ and $C = B$.

Example 5.2

If $f(\mathbf{x}) = B\cdot\mathbf{x} + \mathbf{b}$ with B an n by m matrix and $\mathbf{b} \in \mathbb{R}^n$, then $f(\mathbf{a}+\mathbf{h}) - f(\mathbf{a}) = B\cdot\mathbf{h}$, so the quotient vanishes identically: f is differentiable and $Df(\mathbf{x}) = B$.

We verify that this definition is suitable:

(1) differentiable functions are continuous;

(2) composites of differentiable functions are differentiable (§7);

(3) differentiability at $\mathbf{a}$ implies the existence of all directional derivatives at $\mathbf{a}$.

Theorem 5.1

Let $A \subset \mathbb{R}^m$; let $f : A \to \mathbb{R}^n$. If f is differentiable at $\mathbf{a}$, then all the directional derivatives of f at $\mathbf{a}$ exist, and

$$f'(\mathbf{a};\mathbf{u}) = Df(\mathbf{a})\cdot\mathbf{u}.$$

Proof. Let $B = Df(\mathbf{a})$ and set $\mathbf{h} = t\mathbf{u}$ $(t \neq 0)$ in the definition:

$$\frac{f(\mathbf{a}+t\mathbf{u}) - f(\mathbf{a}) - B\cdot t\mathbf{u}}{|t\mathbf{u}|} \to \mathbf{0} \quad \text{as } t \to 0. \tag{$*$}$$

If $t \to 0$ through positive values, multiply $(*)$ by $|\mathbf{u}|$ to get $\frac{f(\mathbf{a}+t\mathbf{u})-f(\mathbf{a})}{t} - B\cdot\mathbf{u} \to \mathbf{0}$. If $t \to 0$ through negative values, multiply by $-|\mathbf{u}|$ for the same conclusion. Thus $f'(\mathbf{a};\mathbf{u}) = B\cdot\mathbf{u}$. ■

Example 5.3

Define $f : \mathbb{R}^2 \to \mathbb{R}$ by $f(\mathbf{0}) = 0$ and $f(x,y) = x^2y/(x^4+y^2)$ for $(x,y) \neq \mathbf{0}$. All directional derivatives exist at $\mathbf{0}$, but f is not differentiable there. For $\mathbf{u} = (h,k) \neq \mathbf{0}$,

$$\frac{f(\mathbf{0}+t\mathbf{u}) - f(\mathbf{0})}{t} = \frac{(th)^2(tk)}{(th)^4+(tk)^2}\cdot\frac{1}{t} = \frac{h^2k}{t^2h^4+k^2}, \qquad \text{so} \qquad f'(\mathbf{0};\mathbf{u}) = \begin{cases} h^2/k & k \neq 0, \\ 0 & k = 0. \end{cases}$$

If f were differentiable at $\mathbf{0}$ with $Df(\mathbf{0}) = [a\ b]$, then $f'(\mathbf{0};\mathbf{u}) = ah + bk$ would be linear in $\mathbf{u}$, which it is not. In fact f is not even continuous at $\mathbf{0}$: $f(\mathbf{0}) = 0$, while $f(t,t^2) = 1/2$ for all $t \neq 0$. (On each straight line through the origin f is differentiable: on $y = mx$, $f = mx/(m^2+x^2)$.)

Theorem 5.2

Let $A \subset \mathbb{R}^m$; let $f : A \to \mathbb{R}^n$. If f is differentiable at $\mathbf{a}$, then f is continuous at $\mathbf{a}$.

Proof. Let $B = Df(\mathbf{a})$. For $\mathbf{h}$ near $\mathbf{0}$, $\mathbf{h} \neq \mathbf{0}$,

$$f(\mathbf{a}+\mathbf{h}) - f(\mathbf{a}) = |\mathbf{h}|\Big[\frac{f(\mathbf{a}+\mathbf{h}) - f(\mathbf{a}) - B\cdot\mathbf{h}}{|\mathbf{h}|}\Big] + B\cdot\mathbf{h}.$$

As $\mathbf{h} \to \mathbf{0}$ the bracket tends to $\mathbf{0}$, $|\mathbf{h}| \to 0$ and $B\cdot\mathbf{h} \to \mathbf{0}$; hence $f(\mathbf{a}+\mathbf{h}) - f(\mathbf{a}) \to \mathbf{0}$. ■

Definition 5.3 (partial derivative)

Let $A \subset \mathbb{R}^m$, $f : A \to \mathbb{R}$. The **jth partial derivative** of f at $\mathbf{a}$ is the directional derivative with respect to $\mathbf{e}_j$, if it exists:

$$D_j f(\mathbf{a}) = \lim_{t\to 0} \frac{f(\mathbf{a}+t\mathbf{e}_j) - f(\mathbf{a})}{t}.$$

If $\phi(t) = f(a_1, \ldots, a_{j-1}, t, a_{j+1}, \ldots, a_m)$, then $D_j f(\mathbf{a}) = \phi'(a_j)$: one treats $x_1, \ldots, x_{j-1}, x_{j+1}, \ldots, x_m$ as constants and differentiates in x_j by the familiar single-variable rules.

Theorem 5.3

Let $A \subset \mathbb{R}^m$; let $f : A \to \mathbb{R}$. If f is differentiable at $\mathbf{a}$, then

$$Df(\mathbf{a}) = [D_1 f(\mathbf{a}) \;\; D_2 f(\mathbf{a}) \;\; \cdots \;\; D_m f(\mathbf{a})].$$

Proof. $Df(\mathbf{a})$ is a 1 by m matrix $[\lambda_1 \;\cdots\; \lambda_m]$. By Theorem 5.1, $D_j f(\mathbf{a}) = f'(\mathbf{a}; \mathbf{e}_j) = Df(\mathbf{a})\cdot\mathbf{e}_j = \lambda_j$. ■

Theorem 5.4

Let $A \subset \mathbb{R}^m$; let $f : A \to \mathbb{R}^n$. Suppose A contains a neighborhood of $\mathbf{a}$. Let $f_i : A \to \mathbb{R}$ be the ith component function of f, so that $f(\mathbf{x}) = (f_1(\mathbf{x}), \ldots, f_n(\mathbf{x}))$ (a column matrix).

(a) f is differentiable at $\mathbf{a}$ if and only if each component function f_i is differentiable at $\mathbf{a}$.

(b) If f is differentiable at $\mathbf{a}$, then its derivative is the n by m matrix whose ith row is the derivative of f_i.

Thus $Df(\mathbf{a})$ is the matrix whose entry in row i and column j is $D_j f_i(\mathbf{a})$.

Proof. For an arbitrary n by m matrix B consider $F(\mathbf{h}) = (f(\mathbf{a}+\mathbf{h}) - f(\mathbf{a}) - B\cdot\mathbf{h})/|\mathbf{h}|$, defined for $0 < |\mathbf{h}| < \epsilon$. Its ith entry is

$$F_i(\mathbf{h}) = \frac{f_i(\mathbf{a}+\mathbf{h}) - f_i(\mathbf{a}) - (\text{row } i \text{ of } B)\cdot\mathbf{h}}{|\mathbf{h}|}.$$

$F(\mathbf{h}) \to \mathbf{0}$ iff each $F_i(\mathbf{h}) \to 0$. Hence B satisfies the definition for f iff for each i the ith row of B satisfies it for f_i. ■

Definition 5.4 (Jacobian matrix)

If the partial derivatives of the components f_i of f exist at $\mathbf{a}$, the matrix with entry $D_j f_i(\mathbf{a})$ in row i, column j is the **Jacobian matrix** of f at $\mathbf{a}$.

If f is differentiable at $\mathbf{a}$, the Jacobian matrix equals $Df(\mathbf{a})$; but the Jacobian can exist without f being differentiable (Example 5.3). We do not yet have a convenient test for differentiability; e.g. $\sin(xy)$ and xy^2+ze^{xy} obviously have partial derivatives, but are they differentiable? This is settled in §6.

Remark

If $m=1$ or $n=1$, the definition is a matrix reformulation of familiar notions. For $f:\mathbb{R}^1\to\mathbb{R}^3$, $Df(t)$ is the column matrix $(f_1'(t),f_2'(t),f_3'(t))$, the *velocity vector* of the *parametrized-curve* f. For $g:\mathbb{R}^3\to\mathbb{R}^1$, $Dg(\mathbf{x})=[D_1g\ D_2g\ D_3g]$; the vector field $\operatorname{grad} g=(D_1g)\mathbf{e}_1+(D_2g)\mathbf{e}_2+(D_3g)\mathbf{e}_3$ (also written ∇g) is the *gradient*, and the directional derivative $Dg(\mathbf{x})\cdot\mathbf{u}$ is the dot product of $\operatorname{grad} g$ and $\mathbf{u}$. For general $f:\mathbb{R}^m\to\mathbb{R}^n$ matrix notation is needed.

§6. Continuously Differentiable Functions

Motivation. Partial derivatives can exist without f being differentiable. If they are continuous, however, f is differentiable and its derivative is the matrix of partials, so in practice one checks the class C^1 instead of differentiability itself.

Mere existence of partial derivatives does not imply differentiability; their *continuity* does.

Theorem 6.1 (Mean-value theorem)

If $\phi:[a,b]\to\mathbb{R}$ is continuous at each point of $[a,b]$ and differentiable at each point of (a,b), then there exists a point c of (a,b) such that

$$\phi(b)-\phi(a)=\phi'(c)(b-a).$$

Proof [Supplement]. Let $\psi(t)=\phi(t)-\phi(a)-\frac{\phi(b)-\phi(a)}{b-a}(t-a)$; then $\psi(a)=\psi(b)=0$. By the extreme-value theorem ψ attains a maximum and a minimum on $[a,b]$; if both occur at endpoints, $\psi\equiv 0$ and any c works. Otherwise one occurs at some $c\in(a,b)$, where $\psi'(c)=0$ (the difference quotients have opposite signs on the two sides of a local extremum). So $\phi'(c)=(\phi(b)-\phi(a))/(b-a)$. ■

We usually apply it when ϕ is differentiable on an open interval containing $[a,b]$.

Theorem 6.2

Let A be open in $\mathbb{R}^m$. Suppose that the partial derivatives $D_jf_i(\mathbf{x})$ of the component functions of f exist at each point $\mathbf{x}$ of A and are continuous on A. Then f is differentiable at each point of A.

Definition 6.1 (class C^1)

A function satisfying the hypotheses of Theorem 6.2 is **continuously differentiable**, or **of class C^1**, on A.

Proof. By Theorem 5.4 it suffices to treat a real-valued $f:A\to\mathbb{R}$. Let $\mathbf{a}\in A$; the partials $D_jf(\mathbf{x})$ exist and are continuous for $|\mathbf{x}-\mathbf{a}|<\epsilon$.

Step 1. Let $0<|\mathbf{h}|<\epsilon$ with components $h_1,\dots,h_m$, and set

$$\mathbf{p}_0=\mathbf{a},\quad \mathbf{p}_1=\mathbf{a}+h_1\mathbf{e}_1,\quad \mathbf{p}_2=\mathbf{a}+h_1\mathbf{e}_1+h_2\mathbf{e}_2,\quad \dots,\quad \mathbf{p}_m=\mathbf{a}+\mathbf{h}.$$

All $\mathbf{p}_j$ lie in the closed cube C of radius $|\mathbf{h}|$ centered at $\mathbf{a}$. Write

$$f(\mathbf{a}+\mathbf{h}) - f(\mathbf{a}) = \sum_{j=1}^{m} [f(\mathbf{p}_j) - f(\mathbf{p}_{j-1})]. \tag{$*$}$$

Fix j and let $\phi(t) = f(\mathbf{p}_{j-1} + t\mathbf{e}_j)$. Suppose $h_j \neq 0$. As t ranges over the closed interval I with endpoints 0 and h_j, the point $\mathbf{p}_{j-1} + t\mathbf{e}_j$ ranges over the segment from $\mathbf{p}_{j-1}$ to $\mathbf{p}_j$, which lies in $C \subset A$. Only the jth coordinate varies, so since $D_j f$ exists on A, ϕ is differentiable on an open interval containing I. The mean-value theorem gives $\phi(h_j) - \phi(0) = \phi'(c_j)h_j$ for some c_j between 0 and h_j, i.e.

$$f(\mathbf{p}_j) - f(\mathbf{p}_{j-1}) = D_j f(\mathbf{q}_j) h_j, \tag{$**$}$$

where $\mathbf{q}_j = \mathbf{p}_{j-1} + c_j \mathbf{e}_j \in C$. If $h_j = 0$, $(**)$ holds for any $\mathbf{q}_j \in C$. Hence

$$f(\mathbf{a}+\mathbf{h}) - f(\mathbf{a}) = \sum_{j=1}^{m} D_j f(\mathbf{q}_j) h_j, \qquad \mathbf{q}_j \in C. \tag{$***$}$$

Step 2. Let $B = [D_1 f(\mathbf{a}) \ \cdots \ D_m f(\mathbf{a})]$, so $B \cdot \mathbf{h} = \sum_j D_j f(\mathbf{a}) h_j$. By $(***)$,

$$\frac{f(\mathbf{a}+\mathbf{h}) - f(\mathbf{a}) - B\cdot\mathbf{h}}{|\mathbf{h}|} = \sum_{j=1}^{m} [D_j f(\mathbf{q}_j) - D_j f(\mathbf{a})] \frac{h_j}{|\mathbf{h}|}.$$

As $\mathbf{h} \to \mathbf{0}$, $\mathbf{q}_j \to \mathbf{a}$ (it lies in the cube of radius $|\mathbf{h}|$), so by continuity of the partials each bracket tends to 0, while $|h_j|/|\mathbf{h}| \leq 1$. Hence the expression tends to 0. ■

So the familiar functions of calculus, such as $\sin(xy)$ and $xy^2 + ze^{xy}$, are differentiable: their partials exist and are continuous. In practice we deal only with C^1 functions; differentiable functions that are not C^1 occur rarely.

Definition 6.2 (class C^r, C^∞)

Let $f : A \to \mathbb{R}^n$, A open in $\mathbb{R}^m$. If the partials $D_j f_i$ exist on A, their partials $D_k(D_j f_i)$ are the **second-order partial derivatives**; similarly for order r. f is **of class** C^r on A if all partial derivatives of the f_i of order $\leq r$ exist and are continuous on A; equivalently, each $D_j f_i$ is of class C^{r-1}. f is **of class** C^∞ if the partials of all orders are continuous.

Theorem 6.3

Let A be open in $\mathbb{R}^m$; let $f : A \to \mathbb{R}$ be a function of class C^2. Then for each $\mathbf{a} \in A$,

$$D_k D_j f(\mathbf{a}) = D_j D_k f(\mathbf{a}).$$

Proof. Only x_k and x_j vary, so we may assume A open in $\mathbb{R}^2$ and $f : A \to \mathbb{R}$ of class C^2, and prove $D_2 D_1 f = D_1 D_2 f$.

Step 1. A "second-order" mean-value theorem. Let $Q = [a, a+h] \times [b, b+k]$ be a rectangle in A and

$$\lambda(h,k) = f(a,b) - f(a+h,b) - f(a,b+k) + f(a+h,b+k),$$

the sum, with signs, of the values of f at the four vertices. We show there are $\mathbf{p}, \mathbf{q} \in Q$ with

$$\lambda(h,k) = D_2 D_1 f(\mathbf{p}) \cdot hk \qquad \text{and} \qquad \lambda(h,k) = D_1 D_2 f(\mathbf{q}) \cdot hk.$$

By symmetry we prove the first. Let $\phi(s) = f(s, b+k) - f(s,b)$; then $\phi(a+h) - \phi(a) = \lambda(h,k)$. Since $D_1 f$ exists, ϕ is differentiable on an open interval containing $[a, a+h]$, and the mean-value theorem gives s_0 between a and $a+h$ with

$$\lambda(h,k) = [D_1 f(s_0, b+k) - D_1 f(s_0, b)] \cdot h.$$

Now $t \mapsto D_1 f(s_0, t)$ is differentiable on an open interval about $[b, b+k]$ since $D_2 D_1 f$ exists, so by the mean-value theorem again $D_1 f(s_0, b+k) - D_1 f(s_0, b) = D_2 D_1 f(s_0, t_0) \cdot k$ for some t_0 between b and $b+k$. Take $\mathbf{p} = (s_0, t_0)$.

Step 2. Given $\mathbf{a} = (a, b) \in A$ and $t > 0$ small, $Q_t = [a, a+t] \times [b, b+t] \subset A$, and Step 1 gives $\lambda(t,t) = D_2 D_1 f(\mathbf{p}_t)\, t^2$ with $\mathbf{p}_t \in Q_t$. As $t \to 0$, $\mathbf{p}_t \to \mathbf{a}$, and *because $D_2 D_1 f$ is continuous*, $\lambda(t,t)/t^2 \to D_2 D_1 f(\mathbf{a})$. Similarly, using the other equation of Step 1, $\lambda(t,t)/t^2 \to D_1 D_2 f(\mathbf{a})$. The theorem follows. ■

§7. The Chain Rule

Motivation. Almost every computation in the book composes maps: coordinates with functions, parametrizations with forms. The chain rule says the derivative of a composite is the product of the derivative matrices, which is what makes the linear-algebra viewpoint effective.

Theorem 7.1

Let $A \subset \mathbb{R}^m$; let $B \subset \mathbb{R}^n$. Let $f : A \to \mathbb{R}^n$ and $g : B \to \mathbb{R}^p$, with $f(A) \subset B$. Suppose $f(\mathbf{a}) = \mathbf{b}$. If f is differentiable at $\mathbf{a}$, and if g is differentiable at $\mathbf{b}$, then the composite function $g \circ f$ is differentiable at $\mathbf{a}$. Furthermore,

$$D(g \circ f)(\mathbf{a}) = Dg(\mathbf{b}) \cdot Df(\mathbf{a}),$$

where the indicated product is matrix multiplication.

This is the familiar chain rule in a new guise; writing it out in partial derivatives gives the calculus formula (see below).

Proof. Let $\mathbf{x}$ denote the general point of $\mathbb{R}^m$ and $\mathbf{y}$ that of $\mathbb{R}^n$. Choose ϵ so that $g(\mathbf{y})$ is defined for $|\mathbf{y} - \mathbf{b}| < \epsilon$, and δ so that $f(\mathbf{x})$ is defined and $|f(\mathbf{x}) - \mathbf{b}| < \epsilon$ for $|\mathbf{x} - \mathbf{a}| < \delta$ (possible since f is continuous at $\mathbf{a}$). Then $g(f(\mathbf{x}))$ is defined for $|\mathbf{x} - \mathbf{a}| < \delta$.

Step 1. Let $\Delta(\mathbf{h}) = f(\mathbf{a}+\mathbf{h}) - f(\mathbf{a})$ for $|\mathbf{h}| < \delta$. We show $|\Delta(\mathbf{h})|/|\mathbf{h}|$ is bounded in a deleted neighborhood of $\mathbf{0}$. Define $F(\mathbf{0}) = \mathbf{0}$ and $F(\mathbf{h}) = [\Delta(\mathbf{h}) - Df(\mathbf{a}) \cdot \mathbf{h}]/|\mathbf{h}|$ for $0 < |\mathbf{h}| < \delta$. Since f is differentiable at $\mathbf{a}$, F is continuous at $\mathbf{0}$, and

$$\Delta(\mathbf{h}) = Df(\mathbf{a}) \cdot \mathbf{h} + |\mathbf{h}| F(\mathbf{h}) \qquad (*)$$

for $0 < |\mathbf{h}| < \delta$, and trivially for $\mathbf{h} = \mathbf{0}$. By Theorem 1.3 and the triangle inequality, $|\Delta(\mathbf{h})| \le m|Df(\mathbf{a})||\mathbf{h}| + |\mathbf{h}||F(\mathbf{h})|$, and $|F(\mathbf{h})|$ is bounded near $\mathbf{0}$ (it tends to 0).

Step 2. Similarly define $G(\mathbf{0}) = \mathbf{0}$ and $G(\mathbf{k}) = [g(\mathbf{b}+\mathbf{k}) - g(\mathbf{b}) - Dg(\mathbf{b}) \cdot \mathbf{k}]/|\mathbf{k}|$ for $0 < |\mathbf{k}| < \epsilon$. G is continuous at $\mathbf{0}$ and

$$g(\mathbf{b}+\mathbf{k}) - g(\mathbf{b}) = Dg(\mathbf{b}) \cdot \mathbf{k} + |\mathbf{k}| G(\mathbf{k}) \qquad (|\mathbf{k}| < \epsilon). \qquad (**)$$

Step 3. Let $|\mathbf{h}| < \delta$. Then $|\Delta(\mathbf{h})| < \epsilon$, so we may substitute $\mathbf{k} = \Delta(\mathbf{h})$ in $(**)$; since $\mathbf{b} + \Delta(\mathbf{h}) = f(\mathbf{a}+\mathbf{h})$,

$$g(f(\mathbf{a}+\mathbf{h})) - g(f(\mathbf{a})) = Dg(\mathbf{b}) \cdot \Delta(\mathbf{h}) + |\Delta(\mathbf{h})| G(\Delta(\mathbf{h})).$$

Using $(*)$, for $0 < |\mathbf{h}| < \delta$,

$$\frac{1}{|\mathbf{h}|}\big[g(f(\mathbf{a}+\mathbf{h})) - g(f(\mathbf{a})) - Dg(\mathbf{b}) \cdot Df(\mathbf{a}) \cdot \mathbf{h}\big] = Dg(\mathbf{b}) \cdot F(\mathbf{h}) + \frac{|\Delta(\mathbf{h})|}{|\mathbf{h}|} G(\Delta(\mathbf{h})).$$

As $\mathbf{h} \to \mathbf{0}$: $F(\mathbf{h}) \to \mathbf{0}$; $G(\Delta(\mathbf{h})) \to \mathbf{0}$, being a composite of functions continuous at $\mathbf{0}$ and vanishing there; and $|\Delta(\mathbf{h})|/|\mathbf{h}|$ is bounded by Step 1. So the right side tends to $\mathbf{0}$. ■

Corollary 7.2

Let A be open in $\mathbb{R}^m$; let B be open in $\mathbb{R}^n$. Let $f : A \to \mathbb{R}^n$ and $g : B \to \mathbb{R}^p$, with $f(A) \subset B$. If f and g are of class C^r, so is the composite function $g \circ f$.

Proof. By the chain rule, $D(g \circ f)(\mathbf{x}) = Dg(f(\mathbf{x})) \cdot Df(\mathbf{x})$ for $\mathbf{x} \in A$. If f, g are C^1, the entries of Dg are continuous on B, so the entries of $Dg(f(\mathbf{x}))$ are continuous on A; the entries of $Df(\mathbf{x})$ are continuous; and the entries of the product are algebraic (polynomial) functions of these, hence continuous. So $g \circ f$ is C^1.

Induction: suppose the result holds for C^{r-1}, and let f, g be C^r. The entries of Dg are C^{r-1} on B, and f is C^{r-1}, so by the induction hypothesis $D_j g_i(f(\mathbf{x}))$ is C^{r-1}. The entries of Df are C^{r-1}. Hence the entries of $Dg(f(\mathbf{x})) \cdot Df(\mathbf{x})$ are C^{r-1} and $g \circ f$ is C^r. If f, g are C^∞ they are C^r for all r, hence so is $g \circ f$. ■

Theorem 7.3 (Mean-value theorem)

Let A be open in $\mathbb{R}^m$; let $f : A \to \mathbb{R}$ be differentiable on A. If A contains the line segment with end points $\mathbf{a}$ and $\mathbf{a} + \mathbf{h}$, then there is a point $\mathbf{c} = \mathbf{a} + t_0\mathbf{h}$ with $0 < t_0 < 1$ of this line segment such that

$$f(\mathbf{a} + \mathbf{h}) - f(\mathbf{a}) = Df(\mathbf{c}) \cdot \mathbf{h}.$$

Proof. Let $\phi(t) = f(\mathbf{a} + t\mathbf{h})$, defined on an open interval about $[0, 1]$. By the chain rule ϕ is differentiable with $\phi'(t) = Df(\mathbf{a}+t\mathbf{h})\cdot\mathbf{h}$. The ordinary mean-value theorem gives $\phi(1)-\phi(0) = \phi'(t_0)$ for some $0 < t_0 < 1$, i.e. $f(\mathbf{a} + \mathbf{h}) - f(\mathbf{a}) = Df(\mathbf{a} + t_0\mathbf{h}) \cdot \mathbf{h}$. ■

In one variable, if $\phi'(x) > 0$ on an open interval, ϕ has an inverse ψ with $\psi'(y) = 1/\phi'(x)$, $y = \phi(x)$. Here we only compute the derivative of an inverse assuming it exists and is differentiable.

Theorem 7.4

Let A be open in $\mathbb{R}^n$; let $f : A \to \mathbb{R}^n$; let $f(\mathbf{a}) = \mathbf{b}$. Suppose that g maps a neighborhood of $\mathbf{b}$ into $\mathbb{R}^n$, that $g(\mathbf{b}) = \mathbf{a}$, and

$$g(f(\mathbf{x})) = \mathbf{x}$$

for all $\mathbf{x}$ in a neighborhood of $\mathbf{a}$. If f is differentiable at $\mathbf{a}$ and if g is differentiable at $\mathbf{b}$, then

$$Dg(\mathbf{b}) = [Df(\mathbf{a})]^{-1}.$$

Proof. Let $i : \mathbb{R}^n \to \mathbb{R}^n$ be the identity, with $Di = I_n$. Since $g(f(\mathbf{x})) = i(\mathbf{x})$ near $\mathbf{a}$, the chain rule gives $Dg(\mathbf{b}) \cdot Df(\mathbf{a}) = I_n$. So $Dg(\mathbf{b})$ is a left inverse of the square matrix $Df(\mathbf{a})$, hence its inverse (Theorem 2.5). ■

Thus for a differentiable f to have a differentiable inverse it is *necessary* that Df be non-singular. Remarkably, for C^1 functions this is also *sufficient*, at least locally (§8).

Remark on notation

In Leibniz notation, with $y = \phi(x)$, $z = \psi(y)$, the chain rule reads $\frac{dz}{dx} = \frac{dz}{dy} \cdot \frac{dy}{dx}$ and the inverse rule $\frac{dx}{dy} = 1/\frac{dy}{dx}$—easy to remember, but ambiguous (the letter z denotes different functions on the two sides). For $f : \mathbb{R}^m \to \mathbb{R}^n$, $g : \mathbb{R}^n \to \mathbb{R}$ and $F = g \circ f$, the formula $DF(\mathbf{x}) = Dg(f(\mathbf{x})) \cdot Df(\mathbf{x})$ written out entrywise is

$$D_j F(\mathbf{x}) = \sum_{k=1}^{n} D_k g(f(\mathbf{x}))\, D_j f_k(\mathbf{x}), \qquad \text{i.e.} \qquad \frac{\partial z}{\partial x_j} = \sum_{k=1}^{n} \frac{\partial z}{\partial y_k} \frac{\partial y_k}{\partial x_j}$$

with $\mathbf{y} = f(\mathbf{x})$, $z = g(\mathbf{y})$. For the inverse g of $f : \mathbb{R}^2 \to \mathbb{R}^2$, $Dg(\mathbf{y}) = [Df(\mathbf{x})]^{-1}$ reads

$$\begin{bmatrix} \partial x_1/\partial y_1 & \partial x_1/\partial y_2 \\ \partial x_2/\partial y_1 & \partial x_2/\partial y_2 \end{bmatrix} = \begin{bmatrix} \partial y_1/\partial x_1 & \partial y_1/\partial x_2 \\ \partial y_2/\partial x_1 & \partial y_2/\partial x_2 \end{bmatrix}^{-1},$$

so by the formula for the inverse of a matrix, $\partial x_i/\partial y_j$ is far from the reciprocal of $\partial y_j/\partial x_i$. Matrix notation is the clean way to state these results.

§8. The Inverse Function Theorem

Motivation. A linear map with nonzero determinant is invertible. The inverse function theorem says the same holds locally for a C^r map whose derivative is invertible at a point, and the inverse is again C^r. It is the basic tool for producing coordinate systems and is used to define manifolds.

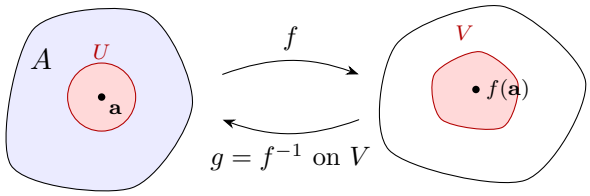


Figure. *The inverse function theorem: if $Df(\mathbf{a})$ is invertible, there is a neighbourhood U of $\mathbf{a}$ that f maps bijectively onto an open set V, and the inverse $g : V \to U$ is again of class C^r. The conclusion is local; f need not be one-to-one on all of A.*

Let A be open in $\mathbb{R}^n$ and $f : A \to \mathbb{R}^n$ of class C^1. We show that non-singularity of Df suffices for f to have a differentiable inverse locally. First, non-singularity implies f is locally one-to-one.

Lemma 8.1

Let A be open in $\mathbb{R}^n$; let $f : A \to \mathbb{R}^n$ be of class C^1. If $Df(\mathbf{a})$ is non-singular, then there exists an $\alpha > 0$ such that the inequality

$$|f(\mathbf{x}_0) - f(\mathbf{x}_1)| \geq \alpha |\mathbf{x}_0 - \mathbf{x}_1|$$

holds for all $\mathbf{x}_0, \mathbf{x}_1$ in some open cube $C(\mathbf{a}; \epsilon)$ centered at $\mathbf{a}$. It follows that f is one-to-one on this open cube.

Proof. Let $E = Df(\mathbf{a})$, non-singular. By Theorem 1.3,

$$|\mathbf{x}_0 - \mathbf{x}_1| = |E^{-1} \cdot (E \cdot \mathbf{x}_0 - E \cdot \mathbf{x}_1)| \leq n|E^{-1}| \cdot |E \cdot \mathbf{x}_0 - E \cdot \mathbf{x}_1|.$$

Setting $2\alpha = 1/(n|E^{-1}|)$, we get $|E \cdot \mathbf{x}_0 - E \cdot \mathbf{x}_1| \geq 2\alpha |\mathbf{x}_0 - \mathbf{x}_1|$ for all $\mathbf{x}_0, \mathbf{x}_1 \in \mathbb{R}^n$.

Let $H(\mathbf{x}) = f(\mathbf{x}) - E \cdot \mathbf{x}$. Then $DH(\mathbf{x}) = Df(\mathbf{x}) - E$, so $DH(\mathbf{a}) = 0$; as H is C^1, choose $\epsilon > 0$ with $|DH(\mathbf{x})| < \alpha/n$ for $\mathbf{x} \in C = C(\mathbf{a}; \epsilon)$. For $\mathbf{x}_0, \mathbf{x}_1 \in C$ (a convex set) the mean-value theorem applied to the ith component of H gives $\mathbf{c} \in C$ with

$$|H_i(\mathbf{x}_0) - H_i(\mathbf{x}_1)| = |DH_i(\mathbf{c}) \cdot (\mathbf{x}_0 - \mathbf{x}_1)| \leq n(\alpha/n)|\mathbf{x}_0 - \mathbf{x}_1|.$$

Hence for $\mathbf{x}_0, \mathbf{x}_1 \in C$,

$$\begin{aligned} \alpha|\mathbf{x}_0 - \mathbf{x}_1| &\geq |H(\mathbf{x}_0) - H(\mathbf{x}_1)| = |f(\mathbf{x}_0) - E \cdot \mathbf{x}_0 - f(\mathbf{x}_1) + E \cdot \mathbf{x}_1| \\ &\geq |E \cdot \mathbf{x}_1 - E \cdot \mathbf{x}_0| - |f(\mathbf{x}_1) - f(\mathbf{x}_0)| \geq 2\alpha|\mathbf{x}_1 - \mathbf{x}_0| - |f(\mathbf{x}_1) - f(\mathbf{x}_0)|, \end{aligned}$$

which is the desired inequality. ∎

Theorem 8.2

Let A be open in $\mathbb{R}^n$; let $f : A \to \mathbb{R}^n$ be of class C^r; let $B = f(A)$. If f is one-to-one on A and if $Df(\mathbf{x})$ is non-singular for $\mathbf{x} \in A$, then the set B is open in $\mathbb{R}^n$ and the inverse function $g : B \to A$ is of class C^r.

Proof.

Step 1. If $\phi : A \to \mathbb{R}$ is differentiable and has a local minimum at $\mathbf{x}_0 \in A$, then $D\phi(\mathbf{x}_0) = 0$. Indeed $\phi(\mathbf{x}_0 + t\mathbf{u}) - \phi(\mathbf{x}_0) \geq 0$ for small t, so the quotient defining $\phi'(\mathbf{x}_0; \mathbf{u})$ is ≥ 0 for $t > 0$ and ≤ 0 for $t < 0$; hence $\phi'(\mathbf{x}_0; \mathbf{u}) = 0$, in particular $D_j\phi(\mathbf{x}_0) = 0$ for all j.

Step 2. B is open. Given $\mathbf{b} \in B$, let $\mathbf{a} = f^{-1}(\mathbf{b})$ and choose a rectangle $Q \subset A$ with $\mathbf{a} \in \operatorname{Int} Q$. $\operatorname{Bd} Q$ is compact, so $f(\operatorname{Bd} Q)$ is compact, hence closed and bounded; since f is one-to-one, $\mathbf{b} \notin f(\operatorname{Bd} Q)$. Choose $\delta > 0$ with $B(\mathbf{b}; 2\delta)$ disjoint from $f(\operatorname{Bd} Q)$. Given $\mathbf{c} \in B(\mathbf{b}; \delta)$ we show $\mathbf{c} = f(\mathbf{x})$ for some $\mathbf{x} \in Q$; then $B(\mathbf{b}; \delta) \subset f(A) = B$.

Let $\phi(\mathbf{x}) = \|f(\mathbf{x}) - \mathbf{c}\|^2$, of class C^r. Since Q is compact, ϕ attains a minimum on Q at some $\mathbf{x}$. Now $\phi(\mathbf{a}) = \|\mathbf{b} - \mathbf{c}\|^2 < \delta^2$, so the minimum is $< \delta^2$. It cannot occur on $\operatorname{Bd} Q$: there $f(\mathbf{x}) \notin B(\mathbf{b}; 2\delta)$, so $\|f(\mathbf{x}) - \mathbf{c}\| \geq \delta$. So $\mathbf{x} \in \operatorname{Int} Q$ and ϕ has a local minimum at $\mathbf{x}$; by Step 1, $D\phi(\mathbf{x}) = 0$. Since $\phi(\mathbf{x}) = \sum_k (f_k(\mathbf{x}) - c_k)^2$, $D_j\phi(\mathbf{x}) = \sum_k 2(f_k(\mathbf{x}) - c_k)D_j f_k(\mathbf{x})$, i.e.

$$2[(f_1(\mathbf{x}) - c_1) \ \cdots \ (f_n(\mathbf{x}) - c_n)] \cdot Df(\mathbf{x}) = 0.$$

$Df(\mathbf{x})$ is non-singular, so multiplying on the right by its inverse gives $f(\mathbf{x}) - \mathbf{c} = \mathbf{0}$.

Step 3. $g : B \to A$ is continuous: for U open in A, $g^{-1}(U) = f(U)$, which is open in $\mathbb{R}^n$ (hence in B) by Step 2 applied to U.

Step 4. g is differentiable at each $\mathbf{b} \in B$. Let $\mathbf{a} = g(\mathbf{b})$, $E = Df(\mathbf{a})$; we show

$$G(\mathbf{k}) = \frac{g(\mathbf{b} + \mathbf{k}) - g(\mathbf{b}) - E^{-1} \cdot \mathbf{k}}{|\mathbf{k}|} \to \mathbf{0} \quad \text{as } \mathbf{k} \to \mathbf{0}.$$

Let $\Delta(\mathbf{k}) = g(\mathbf{b} + \mathbf{k}) - g(\mathbf{b})$. First, $|\Delta(\mathbf{k})|/|\mathbf{k}|$ is bounded for small $\mathbf{k} \neq \mathbf{0}$: by Lemma 8.1 there are a neighborhood C of $\mathbf{a}$ and $\alpha > 0$ with $|f(\mathbf{x}_0) - f(\mathbf{x}_1)| \geq \alpha|\mathbf{x}_0 - \mathbf{x}_1|$ on C; $f(C)$ is a neighborhood of $\mathbf{b}$ (Step 2), so choose ϵ with $\mathbf{b} + \mathbf{k} \in f(C)$ for $|\mathbf{k}| < \epsilon$. Taking $\mathbf{x}_0 = g(\mathbf{b} + \mathbf{k})$, $\mathbf{x}_1 = g(\mathbf{b})$ gives $|\mathbf{k}| \geq \alpha|\Delta(\mathbf{k})|$, i.e. $|\Delta(\mathbf{k})|/|\mathbf{k}| \leq 1/\alpha$. Next,

$$G(\mathbf{k}) = -E^{-1} \cdot \left[\frac{\mathbf{k} - E \cdot \Delta(\mathbf{k})}{|\Delta(\mathbf{k})|}\right] \frac{|\Delta(\mathbf{k})|}{|\mathbf{k}|}$$

(using $\Delta(\mathbf{k}) \neq \mathbf{0}$ for $\mathbf{k} \neq \mathbf{0}$, as g is one-to-one). E^{-1} is constant and $|\Delta(\mathbf{k})|/|\mathbf{k}|$ is bounded. Since $\mathbf{b} + \mathbf{k} = f(g(\mathbf{b} + \mathbf{k})) = f(\mathbf{a} + \Delta(\mathbf{k}))$, the bracket equals

$$\frac{f(\mathbf{a} + \Delta(\mathbf{k})) - f(\mathbf{a}) - E \cdot \Delta(\mathbf{k})}{|\Delta(\mathbf{k})|}.$$

As $\mathbf{k} \to \mathbf{0}$, $\Delta(\mathbf{k}) \to \mathbf{0}$ *because g is continuous*, and since f is differentiable at $\mathbf{a}$ with derivative E, the bracket tends to $\mathbf{0}$.

Step 5. g is of class C^r. By Theorem 7.4, $Dg(\mathbf{y}) = [Df(g(\mathbf{y}))]^{-1}$ for $\mathbf{y} \in B$, so Dg is the composite

$$B \xrightarrow{g} A \xrightarrow{Df} GL(n) \xrightarrow{I} GL(n),$$

where $GL(n)$ is the set of non-singular n by n matrices and $I(C) = C^{-1}$. By Theorem 2.14 the entries of $I(C)$ are rational functions of the entries of C, so I is C^∞. If f is C^1, Df is continuous, g is continuous (even differentiable), so Dg is continuous and g is C^1. If the result holds for C^{r-1} and f is C^r, then g is C^{r-1} by induction, Df is C^{r-1}, so by Corollary 7.2 Dg is C^{r-1} and g is C^r. ■

Remark

Steps 2 and 3 hold without assuming Df non-singular or even f differentiable: if A is open in $\mathbb{R}^n$ and $f : A \to \mathbb{R}^n$ is continuous and one-to-one, then $f(A)$ is open and f^{-1} is continuous (*Brouwer's theorem on invariance of domain*). Its proof requires algebraic topology; we have proved the differentiable version.

Theorem 8.3 (The inverse function theorem)

Let A be open in $\mathbb{R}^n$; let $f : A \to \mathbb{R}^n$ be of class C^r. If $Df(\mathbf{x})$ is non-singular at the point $\mathbf{a}$ of A, there is a neighborhood U of the point $\mathbf{a}$ such that f carries U in a one-to-one fashion onto an open set V of $\mathbb{R}^n$ and the inverse function is of class C^r.

Proof. By Lemma 8.1 there is a neighborhood U_0 of $\mathbf{a}$ on which f is one-to-one. Since $\det Df(\mathbf{x})$ is continuous in $\mathbf{x}$ and $\det Df(\mathbf{a}) \neq 0$, there is a neighborhood U_1 of $\mathbf{a}$ on which $\det Df(\mathbf{x}) \neq 0$. Let $U = U_0 \cap U_1$; the hypotheses of Theorem 8.2 hold for $f : U \to \mathbb{R}^n$. ■

This is the strongest general statement: non-singularity of Df on A makes f locally one-to-one at each point, but not one-to-one on all of A.

Example 8.1

Let $f(r, \theta) = (r\cos\theta, r\sin\theta)$. Then

$$Df(r,\theta) = \begin{bmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{bmatrix}, \qquad \det Df(r,\theta) = r.$$

On the open set $A = (0,1) \times (0,b)$ of the (r,θ)-plane, Df is non-singular at every point, but f is one-to-one on A only if $b \leq 2\pi$.

§9. The Implicit Function Theorem

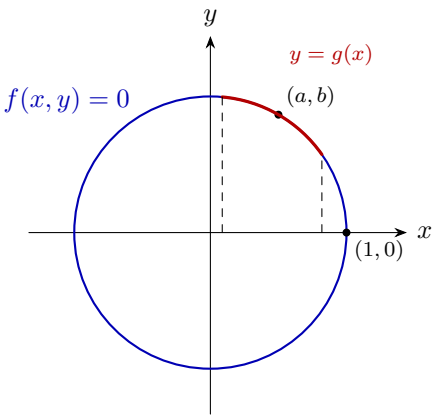


Figure. *Near (a,b) on the circle $x^2+y^2=1$, where $\partial f/\partial y = 2b \neq 0$, the curve is the graph of a C^1 function $y = g(x)$ (red). At $(1,0)$, where $\partial f/\partial y = 0$, no such function exists: every neighbourhood contains two points of the curve over the same x.*

A typical calculus problem: "assume $x^3y + 2e^{xy} = 0$ determines y as a differentiable function of x; find dy/dx." Differentiating, $3x^2y + x^3\,dy/dx + 2e^{xy}(y + x\,dy/dx) = 0$, which one solves for dy/dx. In general, if $f(x,y) = 0$ determines $y = g(x)$ differentiably, then $f(x, g(x)) = 0$ and the chain rule

gives

$$\partial f/\partial x + (\partial f/\partial y)\, g'(x) = 0, \qquad g'(x) = -\frac{\partial f/\partial x}{\partial f/\partial y},$$

evaluated at $(x, g(x))$. This requires $\partial f/\partial y \neq 0$; it turns out this is also *sufficient* for y to be locally a differentiable function of x. More generally, for $f : \mathbb{R}^{k+n} \to \mathbb{R}^n$ of class C^1, the vector equation $f(x_1, \ldots, x_{k+n}) = \mathbf{0}$ is n scalar equations in $k+n$ unknowns; one expects to solve for n unknowns in terms of the other k, with differentiable solutions. There are two problems: computing the derivatives of the implicit functions assuming they exist, and proving existence.

Definition 9.1 (notation for submatrices of Df)

Let A be open in $\mathbb{R}^m$ and $f : A \to \mathbb{R}^n$ differentiable with components $f_1, \ldots, f_n$. We sometimes write

$$Df = \frac{\partial(f_1, \ldots, f_n)}{\partial(x_1, \ldots, x_m)} \qquad \text{or} \qquad Df = \partial f/\partial \mathbf{x}.$$

More generally $\dfrac{\partial(f_{i_1}, \ldots, f_{i_k})}{\partial(x_{j_1}, \ldots, x_{j_l})}$ denotes the k by l matrix of entries of Df in rows $i_1, \ldots, i_k$ and columns $j_1, \ldots, j_l$; its entry in row p, column q is $\partial f_{i_p}/\partial x_{j_q}$.

Theorem 9.1

Let A be open in $\mathbb{R}^{k+n}$; let $f : A \to \mathbb{R}^n$ be differentiable. Write f in the form $f(\mathbf{x}, \mathbf{y})$, for $\mathbf{x} \in \mathbb{R}^k$ and $\mathbf{y} \in \mathbb{R}^n$; then Df has the form

$$Df = [\partial f/\partial \mathbf{x} \quad \partial f/\partial \mathbf{y}].$$

Suppose there is a differentiable function $g : B \to \mathbb{R}^n$ defined on an open set B in $\mathbb{R}^k$, such that $f(\mathbf{x}, g(\mathbf{x})) = \mathbf{0}$ for all $\mathbf{x} \in B$. Then for $\mathbf{x} \in B$,

$$\frac{\partial f}{\partial \mathbf{x}}(\mathbf{x}, g(\mathbf{x})) + \frac{\partial f}{\partial \mathbf{y}}(\mathbf{x}, g(\mathbf{x})) \cdot Dg(\mathbf{x}) = \mathbf{0}.$$

Hence if the n by n matrix $\partial f/\partial \mathbf{y}$ is non-singular at $(\mathbf{x}, g(\mathbf{x}))$,

$$Dg(\mathbf{x}) = -\Big[\frac{\partial f}{\partial \mathbf{y}}(\mathbf{x}, g(\mathbf{x}))\Big]^{-1} \cdot \frac{\partial f}{\partial \mathbf{x}}(\mathbf{x}, g(\mathbf{x})).$$

For $n = k = 1$ this is the formula derived above.

Proof. Let $h : B \to \mathbb{R}^{k+n}$, $h(\mathbf{x}) = (\mathbf{x}, g(\mathbf{x}))$. Then $H(\mathbf{x}) = f(h(\mathbf{x})) = \mathbf{0}$ on B, and the chain rule gives

$$\mathbf{0} = DH(\mathbf{x}) = Df(h(\mathbf{x})) \cdot Dh(\mathbf{x}) = \Big[\frac{\partial f}{\partial \mathbf{x}}(h(\mathbf{x})) \;\; \frac{\partial f}{\partial \mathbf{y}}(h(\mathbf{x}))\Big] \cdot \begin{bmatrix} I_k \\ Dg(\mathbf{x}) \end{bmatrix} = \frac{\partial f}{\partial \mathbf{x}}(h(\mathbf{x})) + \frac{\partial f}{\partial \mathbf{y}}(h(\mathbf{x})) \cdot Dg(\mathbf{x}).$$

■

Theorem 9.2 (Implicit function theorem)

Let A be open in $\mathbb{R}^{k+n}$; let $f : A \to \mathbb{R}^n$ be of class C^r. Write f in the form $f(\mathbf{x}, \mathbf{y})$, for $\mathbf{x} \in \mathbb{R}^k$ and $\mathbf{y} \in \mathbb{R}^n$. Suppose that $(\mathbf{a}, \mathbf{b})$ is a point of A such that $f(\mathbf{a}, \mathbf{b}) = \mathbf{0}$ and

$$\det \frac{\partial f}{\partial \mathbf{y}}(\mathbf{a}, \mathbf{b}) \neq 0.$$

Then there is a neighborhood B of $\mathbf{a}$ in $\mathbb{R}^k$ and a unique continuous function $g : B \to \mathbb{R}^n$

such that $g(\mathbf{a}) = \mathbf{b}$ and $f(\mathbf{x}, g(\mathbf{x})) = \mathbf{0}$ for all $\mathbf{x} \in B$. The function g is in fact of class C^r.

Proof. Define $F : A \to \mathbb{R}^{k+n}$ by $F(\mathbf{x}, \mathbf{y}) = (\mathbf{x}, f(\mathbf{x}, \mathbf{y}))$. Then

$$DF = \begin{bmatrix} I_k & 0 \\ \partial f/\partial \mathbf{x} & \partial f/\partial \mathbf{y} \end{bmatrix},$$

and by repeated application of Lemma 2.12, $\det DF = \det \partial f/\partial \mathbf{y}$, which is nonzero at $(\mathbf{a}, \mathbf{b})$. Also $F(\mathbf{a}, \mathbf{b}) = (\mathbf{a}, \mathbf{0})$. By the inverse function theorem there is an open set $U \times V$ about $(\mathbf{a}, \mathbf{b})$ (U open in $\mathbb{R}^k$, V open in $\mathbb{R}^n$) such that (1) F maps $U \times V$ one-to-one onto an open set W of $\mathbb{R}^{k+n}$ containing $(\mathbf{a}, \mathbf{0})$;

(2) the inverse $G : W \to U \times V$ is of class C^r. Since $F(\mathbf{x}, \mathbf{y}) = (\mathbf{x}, f(\mathbf{x}, \mathbf{y}))$, we have $(\mathbf{x}, \mathbf{y}) = G(\mathbf{x}, f(\mathbf{x}, \mathbf{y}))$, so G preserves the first k coordinates, as F does; write

$$G(\mathbf{x}, \mathbf{z}) = (\mathbf{x}, h(\mathbf{x}, \mathbf{z})), \qquad \mathbf{x} \in \mathbb{R}^k,\ \mathbf{z} \in \mathbb{R}^n,$$

with $h : W \to \mathbb{R}^n$ of class C^r.

Existence. Let B be a connected neighborhood of $\mathbf{a}$ in $\mathbb{R}^k$ so small that $B \times \mathbf{0} \subset W$. For $\mathbf{x} \in B$, $(\mathbf{x}, \mathbf{0}) \in W$ and

$$G(\mathbf{x}, \mathbf{0}) = (\mathbf{x}, h(\mathbf{x}, \mathbf{0})), \qquad (\mathbf{x}, \mathbf{0}) = F(\mathbf{x}, h(\mathbf{x}, \mathbf{0})) = (\mathbf{x}, f(\mathbf{x}, h(\mathbf{x}, \mathbf{0}))),$$

so $f(\mathbf{x}, h(\mathbf{x}, \mathbf{0})) = \mathbf{0}$. Set $g(\mathbf{x}) = h(\mathbf{x}, \mathbf{0})$; g is C^r and $f(\mathbf{x}, g(\mathbf{x})) = \mathbf{0}$. Moreover $(\mathbf{a}, \mathbf{b}) = G(\mathbf{a}, \mathbf{0}) = (\mathbf{a}, h(\mathbf{a}, \mathbf{0}))$, so $g(\mathbf{a}) = \mathbf{b}$.

Uniqueness. Let $g_0 : B \to \mathbb{R}^n$ be continuous with $g_0(\mathbf{a}) = \mathbf{b}$ and $f(\mathbf{x}, g_0(\mathbf{x})) = \mathbf{0}$ on B. If g_0 agrees with g at $\mathbf{a}_0 \in B$, it agrees with g near $\mathbf{a}_0$: since g_0 is continuous and $g_0(\mathbf{a}_0) = g(\mathbf{a}_0) \in V$, there is a neighborhood $B_0 \subset B$ of $\mathbf{a}_0$ with $g_0(B_0) \subset V$. For $\mathbf{x} \in B_0$, $f(\mathbf{x}, g_0(\mathbf{x})) = \mathbf{0}$ gives $F(\mathbf{x}, g_0(\mathbf{x})) = (\mathbf{x}, \mathbf{0})$, and since $(\mathbf{x}, g_0(\mathbf{x})) \in U \times V$, $(\mathbf{x}, g_0(\mathbf{x})) = G(\mathbf{x}, \mathbf{0}) = (\mathbf{x}, h(\mathbf{x}, \mathbf{0}))$, so $g_0 = g$ on B_0. Hence $\{\mathbf{x} \in B : |g(\mathbf{x}) - g_0(\mathbf{x})| = 0\}$ is open in B; so is $\{\mathbf{x} \in B : |g(\mathbf{x}) - g_0(\mathbf{x})| > 0\}$, by continuity. The first contains $\mathbf{a}$, and B is connected, so the second is empty. ∎

Nothing is special about the *last* n coordinates. For example, if $f : A \to \mathbb{R}^2$, A open in $\mathbb{R}^5$, is C^r, $f(\mathbf{a}) = \mathbf{0}$, and $\det \frac{\partial f}{\partial (y,u)}(\mathbf{a}) \neq 0$, then near $\mathbf{a}$ one can solve $f(x, y, z, u, v) = \mathbf{0}$ for $y = \phi(x, z, v)$, $u = \psi(x, z, v)$, with

$$\frac{\partial(\phi, \psi)}{\partial(x, z, v)} = -\Big[\frac{\partial f}{\partial(y, u)}\Big]^{-1} \cdot \Big[\frac{\partial f}{\partial(x, z, v)}\Big].$$

Example 9.1

Let $f(x, y) = x^2 + y^2 - 5$. The point $(1, 2)$ satisfies $f = 0$, and both $\partial f/\partial x$ and $\partial f/\partial y$ are nonzero there, so we can solve locally for either variable in terms of the other. Solving for y gives $y = g(x) = [5 - x^2]^{1/2}$. This is not unique near $x = 1$ unless g is required to be continuous: $h(x) = [5 - x^2]^{1/2}$ for $x \geq 1$, $h(x) = -[5 - x^2]^{1/2}$ for $x < 1$ also satisfies the conditions but is not continuous.

Example 9.2

For the same f, the point $(\sqrt{5}, 0)$ satisfies $f = 0$ but $\partial f/\partial y = 0$ there; indeed no neighborhood B of $\sqrt{5}$ allows solving for y in terms of x.

Example 9.3

Let $f(x,y) = x^2 - y^3$. At $(0,0)$, $\partial f/\partial y = 0$, so we do not expect to solve for y near $(0,0)$. But we can, and the solution $y = x^{2/3}$ is unique; however it is not differentiable at $x = 0$.

Example 9.4

Let $f(x,y) = y^2 - x^4$. At $(0,0)$, $\partial f/\partial y = 0$, yet we can solve for y near $(0,0)$ so that the solution is differentiable ($y = x^2$ or $y = -x^2$); but the solution is not unique. At a point such as $(1,1)$,$\partial f/\partial y \neq 0$, so we can solve for y as a continuous function of x near $x = 1$; one can extend to a larger neighborhood, but if it contains 0 the solution is no longer unique.